



PHYSICS CH.13 — MISCELLANEOUS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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3. States of Matter & Particle Motion
4. Newton's Laws of Motion
5. Simple Machines & Mechanical Advantage
6. Friction
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8. Centre of Gravity & Equilibrium
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1. Scalar vs Vector Quantities

- Scalar = sirf magnitude, no direction; examples: speed, distance, mass, time, temperature, energy, work, power, frequency. [Q1571, Q1580, Q1585, Q1590]
- Vector = magnitude + direction; examples: velocity, force, momentum, acceleration, displacement, weight, electric field. [Q1571, Q1580, Q1585]
- Mass scalar hai; weight vector hai (gravity ki direction mein). [Q1571, Q1590]
- Speed scalar hai; velocity vector hai. [Q1580]
- Mass ek object ka constant property hai — ye place se place change nahi hoti; weight, velocity, aur momentum sab change ho sakte hain. [Q1600]

2. Magnetism & Earth's Magnetic Field

- Freely suspended magnet hamesha North-South direction mein align hota hai. [Q1572]
- Magnet ka North pole geographical North ki taraf point karta hai; actually ye Earth's magnetic South pole hai (opposite poles attract). [Q1572]
- Magnetic field lines North to South (external) aur South to North (magnet ke andar) jaati hain. [Q1572]

3. States of Matter & Particle Motion

- Solid mein particles fixed positions par vibrate karte hain; strongest intermolecular forces. [Q1573, Q1581]
- Liquid mein particles slide past each other; definite volume lekin no definite shape. [Q1573]
- Gas mein particles free aur random motion mein hain; weakest intermolecular forces. [Q1573, Q1581]
- Temperature badhane par solid/liquid/gas ke particles ki kinetic energy increase hoti hai. [Q1573]
- Particles same temperature par different amounts of KE rakhte hain (Maxwell-Boltzmann distribution). [Q1573]

4. Newton's Laws of Motion

- First Law (Inertia): Object rest mein rahega ya uniform motion mein rahega jab tak external force na lage. [Q1575, Q1583]
- Inertia = object ka resistance to change in motion; mass measure of inertia hai — jitna zyada mass, utni zyada inertia. [Q1575, Q1583, Q1595]
- Second Law: $F = ma$; force proportional to acceleration. [Q1575]
- Third Law: Har action ka equal aur opposite reaction. [Q1575, Q1586]
- Rocket propulsion Newton ke Third Law par based hai. [Q1586]
- Jerk lagaane par piche girna First Law (Inertia) ka example hai. [Q1583]
- Acceleration = velocity change per unit time; vector quantity hai; positive ya negative dono ho sakti hai. [Q1589]

5. Simple Machines & Mechanical Advantage

- Mechanical Advantage (MA) = Load/Effort; Velocity Ratio (VR) = Effort distance/Load distance. [Q1574, Q1578]
- Efficiency = $(MA/VR) \times 100\% = (\text{Output/Input}) \times 100\%$. [Q1574, Q1578]
- Lever ka MA = Effort arm / Load arm. [Q1574]
- Inclined plane ka VR = Length/Height; Pulley ka VR = Number of pulleys. [Q1578]
- Efficiency kabhi 100% se zyada nahi hoti; ideal machine mein MA = VR. [Q1574]

6. Friction

- Friction hamesha motion ka virodh karta hai; direction of motion ke opposite. [Q1576, Q1587]
- Static friction > Kinetic friction > Rolling friction. [Q1576]
- Friction reduce karne ke tarike: lubrication, polishing, ball bearings, streamlining. [Q1576]
- Friction increase karne ke tarike: rough surfaces, treads on tyres. [Q1576]
- Friction ke bina chala nahi ja sakta — necessary evil hai. [Q1587]
- Friction ka work lost nahi hota — heat energy mein transfer ho jaata hai (temperature difference ki wajah se). [Q1599]

7. Pressure in Fluids & Archimedes' Principle

- Pressure formula: $P = F/A$; unit Pascal (Pa). [Q1577, Q1582]

- Liquid pressure: $P = h\rho g$; depth ke saath badhta hai. [Q1577]
- Same depth par liquid pressure same hota hai chahe container ka shape jo bhi ho. [Q1577]
- Hydraulic press Pascal's Law par based hai; small force se large force generate hoti hai. [Q1582]
- Archimedes' Principle: Fluid mein submerged object par upward buoyant force = displaced fluid ka weight = $V\rho g$. [Q1579, Q1584]
- Object float karega agar uska density liquid se kam hai; sink karega agar zyada hai. [Q1584]
- Ship steel se bani hai (density > water) lekin hollow structure ki wajah se average density kam ho jaati hai. [Q1584]
- Lactometer milk ki purity check karne ke liye — Archimedes principle. [Q1579]

8. Centre of Gravity & Equilibrium

- Centre of Gravity (CG) = point jahan total weight concentrate hota hai. [Q1588]
- Stable equilibrium: CG neeche, body return karegi original position. [Q1588]
- Unstable equilibrium: CG upar, body gir jaayegi. [Q1588]
- Neutral equilibrium: CG same height, body new position mein rahegi. [Q1588]
- Circus artist CG lower karke balance maintain karta hai. [Q1588]

9. Electromagnetic Induction & Electric Current

- Faraday's Law: Changing magnetic flux se EMF induce hota hai. [Q1601]
- Fleming's Right Hand Rule = generator/dynamo ke liye; thumb = motion, forefinger = field, middle finger = current. [Q1601]
- Fleming's Left Hand Rule = motor ke liye; thumb = force/motion, forefinger = field, middle finger = current. [Q1601]
- Lenz's Law: Induced current ki direction hamesha us change ka virodh karti hai jo usse produce karta hai. [Q1601]
- Eddy currents = changing magnetic field se induced circular currents; undesirable heat produce karte hain; laminated cores se reduce hote hain. [Q1602]
- AC (Alternating Current): direction periodically change hoti hai; India mein 50 Hz frequency. [Q1603, Q1608]
- DC (Direct Current): direction fixed hoti hai; battery, cell se milti hai. [Q1603]
- Peak voltage = RMS voltage $\times \sqrt{2}$; 220 V RMS ka peak ≈ 311 V. [Q1608]

10. Electric Instruments, Safety & Domestic Supply

- India mein domestic supply: 220 V, 50 Hz AC. [Q1608]
- Ammeter = current measure karne ke liye (series mein lagta hai). [Q1592]
- Voltmeter = potential difference measure karne ke liye (parallel mein lagta hai). [Q1592]
- Galvanometer = small current detect karne ke liye. [Q1592]
- Fuse = excess current se circuit protect karta hai; low melting point ki tin-lead alloy se bana hota hai. [Q1593]
- MCB (Miniature Circuit Breaker) = overload ya short circuit par automatic trip; fuse se better kyunki reset ho sakta hai. [Q1623]
- ELCB/RCCB = earth leakage detect karke trip karta hai; shock prevention. [Q1623]
- Three-pin plug = live, neutral, aur earth pin; safety ke liye. [Q1622]
- Earthing/Grounding = excess current ko ground mein bhejta hai; electric shock se bachata hai. [Q1622]

- Electric bell electromagnet par kaam karta hai. [Q1594]
- Transformer = AC voltage step-up ya step-down karta hai; sirf AC ke liye kaam karta hai. [Q1596]
- Step-up transformer: primary turns < secondary turns; voltage badhta hai. [Q1596]
- Step-down transformer: primary turns > secondary turns; voltage ghatta hai. [Q1596]

11. Semiconductor & Electronics

- Semiconductor: conductivity conductor aur insulator ke beech; examples: Silicon (Si), Germanium (Ge). [Q1604]
- Doping se semiconductor ki conductivity badhti hai. [Q1604]
- n-type semiconductor: pentavalent impurity (e.g., Arsenic) add karte hain; electrons majority carriers. [Q1604]
- p-type semiconductor: trivalent impurity (e.g., Boron) add karte hain; holes majority carriers. [Q1604]
- p-n junction diode: current sirf ek direction mein flow hoti hai; rectifier banata hai. [Q1605]
- LED (Light Emitting Diode): forward biased p-n junction se light emit hoti hai. [Q1606]
- Transistor: amplifier ya switch ke roop mein use hota hai; 3 layers (npn ya pnp). [Q1607]
- IC (Integrated Circuit) = chip par multiple transistors/components. [Q1607]

12. Optical Instruments & Phenomena

- Periscope: do parallel plane mirrors 45° par; submerged submarine mein use hota hai. [Q1609]
- Kaleidoscope: multiple plane mirrors; beautiful patterns banate hain. [Q1609]
- Simple microscope: convex lens of short focal length; magnified virtual image. [Q1610]
- Compound microscope: do convex lenses (objective + eyepiece); high magnification. [Q1610]
- Astronomical telescope: distant objects dekhne ke liye; objective large focal length. [Q1610]
- Terrestrial telescope: erect image ke liye; additional lens lagti hai. [Q1610]
- Total Internal Reflection (TIR): light denser se rarer medium mein jaaye aur angle > critical angle; examples: mirage, diamond sparkle, optical fiber. [Q1611]
- Critical angle: $\sin C = 1/\mu$ (μ = refractive index). [Q1611]
- Optical fiber: TIR principle par kaam karta hai; medical endoscopy, communication mein use. [Q1611, Q1628]

13. Nuclear & Atomic Physics

- Isotopes: same atomic number, different mass number; examples: H, D, T; U-235, U-238. [Q1612]
- Isobars: same mass number, different atomic number; examples: Ar-40, Ca-40. [Q1612]
- Isotones: same number of neutrons; examples: C-14, N-15. [Q1612]
- Radioactivity: spontaneous disintegration of unstable nucleus; alpha, beta, gamma emission. [Q1613]
- Half-life: time jisme half of radioactive nuclei decay kar jaate hain. [Q1613]
- Nuclear fission: heavy nucleus split; nuclear reactors, atomic bomb. [Q1613]
- Nuclear fusion: light nuclei combine; sun, stars, hydrogen bomb. [Q1613]
- Nucleus ke andar neutron, proton par jo force lagata hai use nuclear force kehte hain — gravitational, electrostatic, ya tidal force nahi. [Q1597]

14. Gravitation, Satellite & Universe

- Escape velocity: minimum velocity jo object ko Earth's gravity se free kar de; 11.2 km/s. [Q1614]
- Orbital velocity: satellite ki velocity circular orbit mein; ≈ 8 km/s near Earth. [Q1614]
- Geostationary satellite: Earth ke saath rotate karta hai; height $\approx 35,786$ km; period = 24 hours. [Q1614]
- Polar satellite: North-South pole ke charon ore orbit karta hai; Earth mapping ke liye. [Q1614]
- Weightlessness in satellite: free-fall condition, not absence of gravity. [Q1614]
- Free fall sirf vacuum mein hi possible hota hai — air/environment mein resistance ki wajah se object ki acceleration aur velocity affect hoti hai. [Q1598]
- Big Bang Theory: universe ki origin; ~ 13.8 billion years ago. [Q1615]
- Red shift: distant galaxies move away; universe expand ho raha hai. [Q1615]
- Hubble's Law: recession velocity $\propto$ distance; $v = H_0 \times d$. [Q1615]
- Black hole: gravity so strong ki light bhi nahi nikal sakti; event horizon boundary. [Q1616]
- Neutron star: supernova ke baad bacha core; extremely dense. [Q1616]
- Supernova: massive star ka explosion; heavy elements create hote hain. [Q1616]
- Light year: distance light travels in 1 year; $\approx 9.46 \times 10^{12}$ km. [Q1617]
- Astronomical Unit (AU): Earth-Sun average distance; $\approx 1.5 \times 10^8$ km. [Q1617]
- Parsec: larger distance unit; ≈ 3.26 light years. [Q1617]

15. Everyday Physics Applications

- Refrigerator: vapour compression cycle; heat ko fridge se bahar nikalta hai. [Q1618]
- Air conditioner: room se heat bahar nikalta hai; compressor, condenser, evaporator. [Q1618]
- Pressure cooker: high pressure se water ka boiling point badhta hai; $\approx 120^\circ\text{C}$ par cook hota hai. [Q1619]
- Thermos flask (Dewar flask): vacuum between double walls; conduction, convection, radiation sab minimize. [Q1620]
- Sea breeze: day mein land zyada garam, air land se sea ki taraf. [Q1621]
- Land breeze: night mein sea zyada garam, air sea se land ki taraf. [Q1621]
- Monsoon: differential heating se wind patterns; summer mein sea to land. [Q1621]

16. Measurements & Precision Instruments

- Barometer = atmospheric pressure measure karne ke liye. [Q1591]
- Manometer = gas pressure measure karne ke liye. [Q1591]
- Hygrometer = humidity measure karne ke liye. [Q1591]
- Vernier caliper: length measure karne ka precise instrument; least count 0.01 mm ya 0.1 mm. [Q1624]
- Screw gauge: aur bhi precise; least count 0.001 mm ya 0.01 mm. [Q1624]
- Screw gauge ka principle: screw ka pitch; one complete rotation se linear displacement. [Q1624]
- Least count = Pitch / Number of divisions on circular scale. [Q1624]
- Sphygmomanometer = blood pressure measure karne ka instrument. [Q1625]
- Stethoscope = heart/lungs ki sound sunne ke liye. [Q1625]
- Ultrasound (USG) = body ke andar image banane ke liye; >20 kHz. [Q1625]
- ECG (Electrocardiogram) = heart ki electrical activity record karta hai. [Q1625]

- X-ray = haddi fracture dekhne ke liye. [Q1625]

17. Communication & Technology

- Radar: radio waves se object detect karta hai; distance aur speed measure karta hai. [Q1626]
- Sonar: sound waves se underwater object detect karta hai; ships/submarines ke liye. [Q1626]
- LASER: Light Amplification by Stimulated Emission of Radiation; coherent, monochromatic, intense beam. [Q1627]
- LASER uses: surgery, barcode scanner, CD/DVD, cutting, welding. [Q1627]
- Fiber optic communication: TIR principle; high speed data transfer. [Q1628]
- Photocell/Photoelectric effect: light $\rightarrow$ electrical energy; automatic doors, burglar alarms mein. [Q1628]

18. ⚠ Traps / Key Differences Summary (MCQ Q1571–Q1628)

- ⚠ Trap: Mass scalar hai; weight vector hai — confuse mat hona. [Q1571, Q1590]
- ⚠ Trap: Mass constant rehta hai place se place; weight badalta hai gravity ke saath. [Q1600]
- ⚠ Trap: Static friction $>$ Kinetic friction — hamesha. [Q1576]
- ⚠ Trap: Same depth par liquid pressure same hota hai chahe container alag ho. [Q1577]
- ⚠ Trap: Ship steel se bani hai phir bhi float karti hai kyunki average density kam hai. [Q1584]
- ⚠ Trap: Ammeter series mein lagta hai; voltmeter parallel mein. [Q1592]
- ⚠ Trap: Fleming's Right Hand Rule = generator (induction); Left Hand Rule = motor — confuse mat hona. [Q1601]
- ⚠ Trap: Transformer sirf AC ke liye kaam karta hai — DC ke liye nahi. [Q1596, Q1608]
- ⚠ Trap: LED sirf forward biased mein light deta hai; reverse biased mein nahi. [Q1606]
- ⚠ Trap: TIR ke liye light denser se rarer medium mein jaani chahiye aur angle $>$ critical angle hona chahiye. [Q1611]
- ⚠ Trap: Isotopes = same atomic number; Isobars = same mass number; Isotones = same neutrons — teenon alag hain. [Q1612]
- ⚠ Trap: Nuclear force proton-neutron ko bandhta hai — gravitational ya electrostatic force nahi. [Q1597]
- ⚠ Trap: Geostationary satellite ka period 24 hours hota hai; height $\approx$ 35,786 km — fixed position dikhta hai. [Q1614]
- ⚠ Trap: Weightlessness = gravity absence nahi; free-fall condition hai. [Q1614]
- ⚠ Trap: True free fall sirf vacuum mein hoti hai — air resistance ki wajah se normal free fall approximate hi hota hai. [Q1598]
- ⚠ Trap: Light year distance ka unit hai, time ka nahi. [Q1617]
- ⚠ Trap: Pressure cooker mein boiling point badhta hai pressure ke kaaran; normally 100°C se upar. [Q1619]
- ⚠ Trap: Thermos flask mein vacuum hoti hai walls ke beech — conduction aur convection block hote hain. [Q1620]
- ⚠ Trap: MCB fuse se better hai kyunki ye automatic reset ho sakta hai. [Q1623]
- ⚠ Trap: Screw gauge Vernier se zyada precise hota hai; least count chhota hota hai. [Q1624]
- ⚠ Trap: Sonar underwater ke liye; Radar air/land ke liye — donon alag hain. [Q1626]
- ⚠ Trap: Friction ka work lost nahi hota, heat energy mein convert ho jaata hai. [Q1599]
- ⚠ Trap: Inertia jitna zyada mass utni zyada — velocity ya volume par depend nahi karta. [Q1595]

19. Numerical: Relative Density & Density

- Relative density = Density of substance / Density of water; dimensionless (unit nahi hoti). [Q1629, Q1630]
- Metal density = $8.5 \times 10^3 \text{ kg/m}^3$, water = $10^3 \text{ kg/m}^3 \rightarrow$ Relative density = 8.5. [Q1629]
- Silver density = $10.8 \times 10^3 \text{ kg/m}^3$, water = $10^3 \text{ kg/m}^3 \rightarrow$ Relative density = 10.8. [Q1630]
- Density formula: $\rho = m/V$; unit kg/m^3 . [Q1631]
- Mass = 300 kg, Volume = $150 \text{ m}^3 \rightarrow$ Density = $300/150 = 2 \text{ kg/m}^3$. [Q1631]
- Relative density se actual density nikaalne ka formula: Density of substance = Relative density $\times$ Density of water. [Q1632]
- Silver relative density = 10.8 $\rightarrow$ Density = $10.8 \times 10^3 \text{ kg/m}^3$. [Q1632]

20. Numerical: Energy Unit Conversion

- 1 kWh = $3.6 \times 10^6 \text{ J}$; calculation: $1000 \text{ W} \times 3600 \text{ s}$. [Q1633]
- kWh energy ki commercial unit hai — power ki nahi. [Q1633]

21. Numerical: Kinematics — Two-Phase Motion (Rocket)

- Rocket 20 m/s se 35 s tak jaata hai $\rightarrow$ Phase 1 distance = $20 \times 35 = 700 \text{ m}$. [Q1634]
- Fuel cut off ke baad free body ban jaata hai ($u = 20 \text{ m/s}$ upward, $g = 10 \text{ m/s}^2$ neeche). [Q1634]
- Max height tak jaane ke liye: $v^2 = u^2 - 2gs$; $0 = 400 - 20S \rightarrow S = 20 \text{ m}$ extra. [Q1634]
- Total height = $700 + 20 = 720 \text{ m}$. [Q1634]
- ⚠ Phase-wise solve karna hota hai; direct formula apply karne se galat answer aayega. [Q1634]

22. Numerical: Kinematics — Acceleration & Displacement

- Acceleration formula: $a = (v - u)/t$; vector quantity. [Q1635]
- $u = 5 \text{ m/s}$, $v = 10 \text{ m/s}$, $t = 5 \text{ s} \rightarrow a = (10 - 5)/5 = 1 \text{ m/s}^2$. [Q1635]
- Vertical throw mein displacement: $s = ut - (1/2)gt^2$ (upward positive, g negative). [Q1636, Q1637]
- Ball 80 m max height tak jaati hai; u nikaalne ke liye: $v^2 = u^2 - 2gs$; $0 = u^2 - 1600 \rightarrow u = 40 \text{ m/s}$. [Q1636]
- 7 s baad displacement: $s = 40 \times 7 - 0.5 \times 10 \times 49 = 280 - 245 = 35 \text{ m}$. [Q1636]
- $u = 40 \text{ m/s}$, $t = 6 \text{ s}$, $g = 10 \text{ m/s}^2 \rightarrow s = 40 \times 6 - 0.5 \times 10 \times 36 = 240 - 180 = 60 \text{ m}$. [Q1637]

23. Numerical: Kinematics — Vertical Throw Problems

- Time to reach highest point: $v = u - gt$; highest point par $v = 0$. [Q1640]
- $u = 14 \text{ m/s}$, $g = 9.8 \text{ m/s}^2 \rightarrow t = 14/9.8 = 1.43 \text{ s}$. [Q1640]
- ⚠ g ki value exact 9.8 use karni padti hai; 10 se approximate nahi karna jab options close ho. [Q1640]

24. Numerical: Kinematics — Equal Time Intervals under Constant Acceleration

- Rest se start karke constant acceleration mein: $s = ut + (1/2)at^2$; $u = 0$ hone par $s \propto t^2$. [Q1638]
- Pehle 10 s mein distance $X_1 = (1/2) \times a \times 100 = 50a$. [Q1638]
- Velocity at 10 s: $v = u + at = 10a$. [Q1638]
- Agle 10 s (11–20 s) mein distance $X_2 = 10a \times 10 + (1/2) \times a \times 100 = 100a + 50a = 150a$. [Q1638]
- Ratio: $X_2 = 3X_1$; yaani doosre equal interval mein pehle interval se 3 times distance. [Q1638]
- Equal time intervals mein distance ratio from rest: 1 : 3 : 5 : 7 ... (odd numbers). [Q1638]
- Object rest se 4 m/s² se 8 s baad speed: $v = u + at = 0 + 4 \times 8 = 32$ m/s. [Q1639]

25. Numerical: Formula Cheat Sheet

- Relative Density: $RD = \rho_{\text{substance}} / \rho_{\text{water}}$.
- Density: $\rho = m/V$.
- 1 kWh = 3.6×10^6 J.
- Acceleration: $a = (v - u)/t$.
- 1st Equation: $v = u + at$.
- 2nd Equation: $s = ut + (1/2)at^2$.
- 3rd Equation: $v^2 = u^2 + 2as$.
- Vertical throw displacement: $s = ut - (1/2)gt^2$.
- Time to max height: $t = u/g$ (at $v=0$).
- Distance ratio (equal intervals, $u=0$): 1 : 3 : 5 : 7 ...

26. ⚠ Traps / Key Differences Summary (Numerical Q1629–Q1640)

- ⚠ Trap: Relative density ka unit nahi hota — ye dimensionless ratio hai. [Q1629, Q1630]
- ⚠ Trap: Density nikaalne ke liye mass ÷ volume karte hain; unit kg/m³ hoti hai. [Q1631]
- ⚠ Trap: Rocket jaise two-phase problems mein pehle constant velocity phase alag solve karo, phir free-fall phase alag. [Q1634]
- ⚠ Trap: Vertical throw mein displacement direction consider karta hai; distance sirf total path length hai. [Q1636, Q1637]
- ⚠ Trap: Time to highest point nikaalne mein $v = 0$ daalna hota hai; g exact 9.8 se divide karo jab options close hon. [Q1640]
- ⚠ Trap: Equal time intervals mein distance ratio 1:3:5... sirf rest se start karne par hota hai; initial velocity ho to alag hoga. [Q1638]
- ⚠ Trap: 1 kWh = 3.6×10^6 J — 3.6×10^5 ya 36×10^6 options confuse karne ke liye hote hain. [Q1633]